BIOCHEMISTRY CUMULATIVE EXAM

February 8, 1997

Vesicles were prepared by sonicating a mixture of phospholipids, a newly discovered ATP synthase, and gramicidin A in a pH 6 buffer containing 5 mM Na^+Cl^-. This procedure results in the incorporation of ATP synthase into the vesicle with the F_1 subunit responsible for the synthesis of ATP oriented toward the outside of the vesicle. The vesicles were then washed with a pH 8 buffer containing 100 mM Na^+Cl^-. After this wash, the vesicles thus contain 5 mM Na^+Cl^- and pH 6 buffer on the inside of the vesicle, and 100 mM Na^+Cl^- and pH 8 buffer on the outside of the vesicle.

A. Calculate the voltage difference between the outside and inside of the vesicle membrane, assuming that the gramicidin A in the membrane allows efficient and selective transport of Na^+ across the membrane.

B. Calculate the free energy ΔG for transporting each mole of protons across the membrane from inside to outside of the vesicle.

C. Calculate the actual free energy ΔG for synthesizing ATP in the solution on the outside of the vesicle, assuming that the outside solution contains the following concentrations: $[ATP] = [P_i] = 10$ mM, $[ADP] = 1$ mM.

D. Determine whether the free energy of proton transport calculated in part B is sufficient to drive the synthesis of ATP by the ATP synthase under the conditions of part C, assuming that the number of protons transported per ATP synthesized is 3. Determine the <u>minimum</u> differenece in pH that would be required to synthesize ATP, assuming the voltage gradient calculated in part A.

Possible Equations:

$\Delta G^{o'}$(ATP synthesis) = + 7.3 KCal/mol.

$\Delta G(X_{out} \rightarrow X_{in}) = RT\ln([X_{in}]/[X_{out}]) + ZF(V_{in} - V_{out})$.

$\Delta G = \Delta G^o + RT\ln([\text{products}]/[\text{reactants}])$.

R = 2 cal/degree.mol F = 23 KCal/volt.mol